

Ziqi Liu

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EDUCATION

• The Hong Kong University of Science and Technology

Sep 2025 – Expected Jun 2027

Master of Philosophy, Integrated Systems and Design

Hong Kong

- **Advisor:** [Qijia Shao](#)
- **Research Focus:** Mobile & Ubiquitous Computing, HCI.
- I focus on wearable and mobile sensing technologies, adaptive interaction support, and mobile health, aiming to design AI-assisted systems that implicitly sense users' behavioral and physiological states and provide timely, context-aware interventions.

• Tsinghua University

Sep 2021 – Jun 2025

Bachelor of Engineering, Intelligent Engineering and Creative Design

Beijing, China

- **Advisors:** [Yuntao Wang](#), [Haipeng Mi](#)
- GPA: 3.70/4.0
- An interdisciplinary undergraduate program that combines Electrical Engineering and Computer Science (EECS, main part), Mechanical Engineering (ME), and Industrial Design (ID).

PUBLICATIONS

- [1] **Ziqi Liu**, Ziyi Xu, Jinhe Wen, Xiangjie Tang, and Qijia Shao. 2026. **PreTap: Implicit Reachability Analysis and Tap Prediction for One-Handed Mobile Interaction**. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (Accepted by IMWUT '26).
- [2] Xiangjie Tang, **Ziqi Liu**, Minhao Cui, and Qijia Shao. 2026. **Turning Phone-Surface Contacts into Probabilistic Landmarks for Indoor Localization**. Submitted to *ACM Conference on Embedded Networked Sensor Systems* (Submitted to SenSys '27).
- [3] Yibo Wang, **Ziqi Liu**, Jiao Xue, and Qi Lu. 2025. **AroMR: Decentralizing Olfactory Displays into the Environment for Olfactory-Augmented Experiences in Mixed Reality**. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (CHI EA '25). [\[Paper\]](#)
- [4] Chang Liu, Xiangyang Wang, Chun Yu, Yingtian Shi, Chongyang Wang, **Ziqi Liu**, Chen Liang, and Yuanchun Shi. 2025. **Enhancing Smartphone Eye Tracking with Cursor-Based Interactive Implicit Calibration**. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). [\[Paper\]](#)

RESEARCH EXPERIENCE

• UbiquitousX Lab, The Hong Kong University of Science and Technology

09/2025 – Now

M.Phil Student / Advisor: Prof. [Qijia Shao](#)

Hong Kong

- Currently investigating how physical pressure, body posture, and related contextual factors affect wrist-worn PPG signals, aiming to support more robust signal recovery, physiological state sensing, and downstream mobile health applications.
- Led projects related to user-state sensing and interaction-assistance on smartphones, including *PreTap* (IMWUT '26) and *STEP* (submitted to SenSys '27), developing on-device AI systems that leverage implicit motion sensing to infer user-specific states and provide personalized interaction support.

• Pervasive HCI Group, Tsinghua University

05/2024 – 06/2025

Undergraduate Research Assistant / Advisor: Prof. Chun Yu, Prof. [Yuntao Wang](#)

Beijing, China

- Conducted algorithm design for COMETIC (CHI '25), which introduced a cursor-based implicit calibration method for smartphone eye tracking that leverages correlations between cursor operations and gaze patterns to continuously fine-tune tracking accuracy during natural interaction.
- Designed a pen-based sensing system that integrates EDA, PPG, and IMU sensors to enable unobtrusive physiological monitoring and cognitive load assessment during natural handwriting interaction.

• Future Lab, Tsinghua University

03/2024 – 01/2025

Undergraduate Research Assistant / Advisor: Prof. [Haipeng Mi](#), Prof. [Qi Lu](#)

Beijing, China

- Co-led AroMR (CHI EA '25), which introduces a field-centric olfactory rendering strategy by decentralizing displays into the environment to create immersive, context-aware scent experiences in Mixed Reality.
- Conducted interaction and system design to explore the redesign possibilities of traditional embroidery using electroluminescent threads, proposing a framework for sustainable cultural preservation through smart materials.

INDUSTRY EXPERIENCE

- **Shokz** 06/2025 – 08/2025
Research Intern Shenzhen, China
 - **Project:** Design and Deployment of Clamping Force Test Sensors for Head-Mounted Headphones
 - Developed clamping force measurement systems and wearable test prototypes for two bone conduction headphone products, and conducted user and wearing tests to support pre-research application scenarios.
- **XIAO MI** 01/2025 – 03/2025
Product Manager Intern, Corporate Group Technology Committee – Xiao Ai Interconnection Group Beijing, China
 - Designed and implemented voice interaction response strategies for scenarios with coexistence of multiple devices such as smartphones, smart home devices, and wearable devices.
- **Mercedes Benz (Beijing) & Tsinghua University** 08/2024 – 11/2024
Research Intern Beijing, China
 - **Project:** Towards Sustainable Car Interior Design with Smart Interactive Material
 - Designed and fabricated the high-fidelity demo for interior design with interactive materials, responsible for lighting effects design and circuit implementation.

TEACHING EXPERIENCE

- **ISDN5230 – Artificial Intelligence of Things for Healthcare** Spring 2026
Teaching Assistant / Instructor: Qijia Shao Hong Kong

ACADEMIC SERVICE

- **ACM IMWUT** 10/2026
Student Volunteer Shanghai, China
- **ACM MobiCom** 11/2025
Student Volunteer Hong Kong

ADVISING AND COLLABORATING

- **Yin Cao**, Undergraduate Student in ECE, University of Florida Present
- **Xiangjie Tang**, Undergraduate Student in CS, Southeast University; now CS Ph.D., SNU 2025 – 2026
- **Ziyi Xu**, Undergraduate Student in ISD, HKUST; now Ph.D., HKUST 2025 – 2026

SKILLS

- **Programming Languages:** Python, C/C++, HTML, shell
- **Embedded Systems & Hardware Design:** Raspberry Pi, ESP32, Arduino, Verilog, QuartusII, Multisim
- **Modeling & Graphic Design:** AutoCAD, Solidworks, Fusion 360, Figma, Adobe suite, Unity, Blender
- **Skill Set:** Machine Learning (PyTorch), User Interface Design, Digital Fabrication, Rapid Prototyping